Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Computer Application
BCA / B.Sc. (IT) | Sem-2 | Summer-2025

Course Code	:	2304CS204	**Date**	:	23-04-2025
Course Name	:	Database Management System - I	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** List and explain applications of DBMS. **4**

(B) Define the following terms: **3**
(i) Data
(ii) Information
(iii) Metadata

OR

Define the following terms:
(i) Field, Record and File
(ii) DBMS
(iii) Data Dictionary

(C) Explain the three level ANSI SPARC Database Architecture. **7**

OR

Explain the advantages of DBMS over a File-oriented System.

Q.2 **(A)** Discuss the following terms: **4**
(i) Schema
(ii) Mapping
(iii) Data Independence

(B) Write a short note on the Hierarchical Data Model. **3**

OR

Write a short note on the Network Data Model.

(C) List different types of Database Systems and explain any two with appropriate diagrams. **7**

OR

Describe the functions and services of a Database Management System.

Q.3 **(A)** Discuss the Components of SQL in brief. **4**

(B) Discuss the ORDER BY and GROUP BY clause with suitable examples. **3**

OR

Describe the ALTER command with proper syntax and an example.

(C) Consider the following table and prepare the given queries: **7**

Car (car_ID, model, year, color, price)

1. Display cars within price range of 3,00,000 to 5,00,000.
2. Find the average car price ordered by model.
3. Update the car price by 10% where car ID is 101.
4. Delete car details where car ID is 105.
5. List the top 4 cars with model, year and price.
6. Alter the data type of price field to Decimal(6,2).
7. Count the white-colored car model-wise.

OR

Consider the following table and prepare the given queries:

Hotel (hotel_ID, hotel_name, city, address, phone_no, email, star_rating)

1. Display all 5-star hotels ordered by hotel name.
2. Display hotel details with ID, name, city and email.
3. Update the hotel star rating to 4 where the hotel name is 'Heritage Villa'.
4. Delete all information about hotels located in Udaipur.
5. Count the 3-star rating hotels where hotel name starts with 'A' and end with 'S'.
6. Convert all hotel names in uppercase and cities in lowercase.
7. Discard the hotel table from the database.

Q.4 **(A)** Write a short note on JOIN operations in relational algebra. **4**

(B) Describe the UNION operation in relational algebra with a proper example. **3**

OR

Describe the INTERSECTION operation in relational algebra with a proper example.

(C) Explain the SELECTION operation in relational algebra with a suitable example. **7**

OR

Explain the PROJECTION operation in relational algebra with a suitable example.

Q.5 **(A)** Prepare an E-R diagram for a Library Management System. Assume suitable entities and attributes. **4**

(B) Discuss different types of attributes in the E-R diagram. **3**

OR

Discuss Weak Entity and Strong entity with a proper example.

(C) Explain Generalization and Specialization with a suitable example. **7**

OR

Explain an E-R diagram with its symbols.

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